FORSYS
Forward Thinking

NextGen Testing powered by Automation

Client Background

The client had no formal product life cycle management and software development life cycle. In addition, they were not able to document and analyze defects, test applications and maintain standards. As a result, it was difficult to measure the overall effectiveness of testing and, in effect, the software quality. The client was looking to establish a quality assurance (QA) structure and process that is capable of meeting the industry requirements.

Solution

Forys used a global delivery model with onsite and offshore. While one resource is still available at the client site, we still implement and manage the QA project from offshore. We established industry-wide best practices in documentation and QA cycle management.

We developed high-level test scenarios and mapped them to test cases, while ensuring that the high level test scenarios are reviewed and signed off from customers in the design phase. Our Business Driven QA approach helped us to identify critical business test scenarios in advance. We measured the testing coverage provided by these high level scenarios and detailed our findings in a report at the end of each testing cycle. We executed 200 test cases and identified 40 critical errors. We implemented the new QA process for one complete software test life cycle (STLC) for the client.

Technical Challenges

During the course of the project, Forsys faced the following challenges:

- **Automation tool selection -** The challenge was to select the GUI automation testing tool and software quality assurance and governance tool. After careful assessment, we recommended Selenium which is Free and Open source tool. We also assessed the potential future risks that the client might encounter by using these tools.
    - Selenium tool benefits
    - Ease of implementation
    - Multi OS/Multi Browser Supports
    - Parallel Test execution
    - Integration with other tools like Test Management, CICD, Defect Tracking etc.
    - Customized automation framework
    - Screen capture and Flexibility
    - Easy to read logs and debug
    - HTML, Customized Reports
    - Easy maintenance

Tool compatibility

Challenges were encountered in determining if JIRA (Bug Tracking Tool) and Test Management Tool were compatible.

Benefits

After implementing Forsys's recommendations for its QA process, the client had the following benefits:

- Established a formal product lifecycle management and software development life cycle.
- Improved the product quality through a QA process and wider coverage of testing areas
- Achieved faster time to market and quicker ROI along with assured quality results through Forsys customized selenium Quality automation framework (QAF)
- Reduced the test cycle time by 20-25% by leveraging Forsys Quality CoE automation framework
- Established industry standard QA practices.